

Lab Guide: Resolving Layer 2 Broadcast Storm

 STOP: SPOILER ALERT!

Have you attempted to solve the simulation on your own at least twice? The Network+ exam doesn't just test what you know; it tests how you think under pressure. If you jump straight to this guide, you are robbing yourself of the "troubleshooting muscle memory" required to pass the PBQ. Try to find the loop first!

1. The "How and Why" of the Problem

In this topology, two physical paths exist between **SW1 (Core)** and **SW2 (Access)**. Without a loop-prevention mechanism, a single broadcast frame will circulate between these switches indefinitely, multiplying until CPU utilization hits 100% and the network becomes unresponsive. This is known as a **Broadcast Storm**.

2. Phase 1: Evidence Collection

Before changing any settings, you must understand the current state of the environment:

- **The Symptom:** Note the "Storm Detected" status. In a real-world scenario, you would observe all port LEDs blinking in rapid, perfect synchronization.
- **The Investigation:** Run show spanning-tree vlan 10 on both switches.
- **The Smoking Gun:** Compare the **Root ID** and **Bridge ID**. If they match on **SW2**, the Access switch has elected itself as the "Root" of the network. This typically occurs because SW2 has a lower MAC address than the Core switch while both are at default priority.

3. Phase 2: The Resolution (The "Why" for SW1)

To fix this, we are implementing **Manual Prioritization**. In a professional network, we never let the hardware decide the topology for us. We want to 'force' the Core switch to be the leader because it's the strongest link in the chain.

Why SW1 MUST be the Root:

In a tiered network, the **Core Switch (SW1)** acts as the central hub. If an **Access Switch (SW2)** is elected as the Root, it creates a massive performance bottleneck known as **sub-optimal pathing**. Consider the following:

Scenario	Impact of SW2 as Root
Centralized Traffic	The Core switch is designed for high-speed throughput. Making the Access switch the Root forces all traffic to flow through weaker hardware.
Network Expansion	If other switches or external networks are connected to the "other side" of the Core, traffic from those segments would have to travel all the way to SW2 just to be redirected back through the Core to reach its destination.
Latency	Forcing traffic through unnecessary "hops" at the access layer increases latency and consumes bandwidth on slower access-layer uplinks.

The Fix: Use the **command spanning-tree vlan 10 root primary** on **SW1**. This forces the switch to lower its **Priority** (typically to 4096), ensuring it wins the election and remains the center of the network universe.

4. Phase 3: Verification

After running the fix, check **SW2** again:

- **Interface G0/2** should now show a status of **Altn BLK** (Alternate/Blocking).
- The "Storm" is now logically broken. Data flows through G0/1, but G0/2 stands ready to take over automatically if the primary link fails.